



**BHARAT PETROLEUM CORPN. LTD.**  
(A Govt. of India Undertaking)  
**QUALITY ASSURANCE LABORATORY**  
**LONI, GHAZIABAD, UTTAR PRADESH - 201102**  
**NABL Certificate. No. TC-7105**

**TEST REPORT**  
**MAK HYDROL AW 68**

|                       |                      |                    |                         |
|-----------------------|----------------------|--------------------|-------------------------|
| Test Report Issued To | PM                   | Date               | Nov 08, 2023            |
| Test Report No        | LON/BF(L)/2311/00525 | Type Of Sample     | Batch Formation (Lubes) |
| Batch No.             | L12444/09            | Sampled Location   | MAK LUBE PLANT LONI     |
| Product Code          | 7662500              | Sample Received On | Nov 08, 2023            |
| Date Of Sampling      | Nov 08, 2023         | Date of Testing    | Nov 08, 2023            |

| S. No. | Characteristics                              | TEST METHOD                                        | Test Result |
|--------|----------------------------------------------|----------------------------------------------------|-------------|
| 1      | Appearance                                   | Visual                                             | Clear       |
| 2      | Colour, ASTM                                 | IS:1448(P-12)/D 1500                               | 0.5         |
| 3      | Crackle Test                                 | ---                                                | Negative    |
| 4      | Density at 29.5 deg C, ASTM D4052            | ASTM D4052                                         | 0.8480      |
| 5      | Pour Point (P:10), deg C                     | P-10                                               | -27 deg C   |
| 6      | Flash Point, COC, Deg C, ASTM D92            | ASTM D92                                           | 224         |
| 7      | Kinematic Viscosity, cSt @ 40 °C, ASTM D445  | ASTM D445                                          | 65.19       |
| 8      | Kinematic Viscosity, cst @ 100 °C, ASTM D445 | ASTM D445                                          | 9.274       |
| 9      | Viscosity Index                              | ASTMD:2270-93 / IS:1448 (P56) 1980 Reaffirmed 2003 | 120         |
| 10     | Emulsion Characteristics (NOT IN NABL SCOPE) | ASTM D1401-02                                      |             |
| 10.1   | Emulsion Characteristics                     | ASTM D1401-02                                      | 40-40-0(20) |

REMARK: THE PRODUCT AS REPRESENTED BY THE SAMPLE MEETS THE SPECIFICATION OF MAK HYDROL AW 68 FOR THE TESTS CARRIED OUT AS PER INTERNAL SPECIFICATION

\*\*\* End of Test Report\*\*\*

Notes:

1. The sample is drawn by the client & result relates to the sample tested.
2. The certificate shall not be reproduced wholly or in part without prior written consent of the laboratory
3. This certificate shall not be used in any advertising media or as evidence in the court of law without written consent of laboratory
- \* Not in NABL Scope
4. Latest version of test methods are used as per latest specification

|                                                                                        |                                |             |             |
|----------------------------------------------------------------------------------------|--------------------------------|-------------|-------------|
| Authorised Signatory                                                                   |                                | Manini.Wagh |             |
| Printed by: Manini.Wagh                                                                | Printed on: Dec 1 2023 12:09PM |             | Copy No.: 1 |
| Note : This document has been generated electronically and is valid without signature. |                                |             |             |
| Format No.QR/69                                                                        |                                |             |             |